Exercise # 1.1

Q3 $x_1 + 5x_2 = 7$ ——— ①

$x_1 - 2x_2 = -2$ ——— ②

eq① − eq②

$$\begin{array}{r} x_1 + 5x_2 = 7 \\ -x_1 \mp 2x_2 = \mp 2 \\ \hline 7x_2 = 9 \end{array}$$

$$\boxed{x_2 = 9/7}$$

Put in eq①

$$x_1 + 5\left(\frac{9}{7}\right) = 7$$

$$x_1 = 7 - \frac{45}{7}$$

$$x_1 = \frac{49-45}{7} = \frac{4}{7}$$

$$\boxed{x_1 = 4/7}$$

Both lines intersect at point $\left(\frac{4}{7}, \frac{9}{7}\right)$

Q6

Replace R_4 by its sum with -3 times R_3.
After this scale the R_4 by $-1/5$

Q10

$$\begin{bmatrix} 1 & -2 & 0 & 3 & -2 \\ 0 & 1 & 0 & -4 & 7 \\ 0 & 0 & 1 & 0 & 6 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix} \sim r_1 \Leftarrow r_1 + 2(r_2)$$

$$\begin{bmatrix} 1 & 0 & 0 & -5 & 12 \\ 0 & 1 & 0 & -4 & 7 \\ 0 & 0 & 1 & 0 & 6 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix} \sim r_1 \Leftarrow r_1 + 5(r_4)$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & -3 \\ 0 & 1 & 0 & -4 & 7 \\ 0 & 0 & 1 & 0 & 6 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix} \quad \tilde{r}_2 \leftarrow 4(r_4) + r_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & -3 \\ 0 & 1 & 0 & 0 & -5 \\ 0 & 0 & 1 & 0 & 6 \\ 0 & 0 & 0 & 1 & -3 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -3 \\ -5 \\ 6 \\ -3 \end{bmatrix} \quad \underline{Ans}$$

Q12 $x_1 - 3x_2 + 4x_3 = -4$

$3x_1 - 7x_2 + 7x_3 = -8$

$-4x_1 + 6x_2 - x_3 = 7$

$$\begin{bmatrix} 1 & -3 & 4 & -4 \\ 3 & -7 & 7 & -8 \\ -4 & 6 & -1 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 & 4 & -4 \\ 0 & 2 & -5 & 4 \\ 0 & -6 & 15 & -9 \end{bmatrix} \quad \begin{matrix} \tilde{r}_2 \leftarrow r_2 - 3r_1 \\ \tilde{r}_3 \leftarrow r_3 + 4r_1 \end{matrix}$$

$$\begin{bmatrix} 1 & -3 & 4 & -4 \\ 0 & 2 & -5 & 4 \\ 0 & 0 & 0 & 3 \end{bmatrix} \quad \tilde{r}_3 \leftarrow r_3 + 3r_2$$

In augmented Matrix form:-

$$\begin{bmatrix} 1 & 0 & 0 & -2 & -3 \\ 0 & 2 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ -2 & 3 & 2 & 1 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & -2 & -3 \\ 0 & 2 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 3 & 2 & -3 & -1 \end{bmatrix} \quad \tilde{r}_4 \leftarrow r_4 + 2r_1$$

$$\begin{bmatrix} 1 & 0 & 0 & -2 & -3 \\ 0 & 2 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & -1 & -3 & -1 \end{bmatrix} \quad \tilde{r}_4 \leftarrow r_4 + \left(\frac{-3}{2}\right) r_2$$

$$\begin{bmatrix} 1 & 0 & 0 & -2 & -3 \\ 0 & 2 & 2 & 0 & 0 \\ 0 & 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \tilde{r}_4 \leftarrow r_3 + r_4$$

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$$\begin{bmatrix} 1 & h & 4 \\ 3 & 6 & 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & h & 4 \\ 0 & 6-3h & -4 \end{bmatrix} \quad \tilde{r}_2 \leftarrow r_2 + (-3) r_1$$

As, the system is consistent, so, $h \neq 2$

Q25

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ -2 & 5 & -9 & k \end{bmatrix}$$

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ 0 & -3 & 5 & k+2g \end{bmatrix} \sim r_3 \leftarrow r_3 + 2r_1$$

$$\begin{bmatrix} 1 & -4 & 7 & g \\ 0 & 3 & -5 & h \\ 0 & 0 & 0 & h+k+2g \end{bmatrix} \sim r_3 \leftarrow r_3 + r_2$$

Q33

$$4T_1 - T_2 - T_4 = 30$$
$$4T_2 - T_1 - T_3 = 60$$
$$4T_3 - T_4 - T_2 = 70$$
$$4T_4 - T_1 - T_3 = 40$$

Q34 By rearranging the equations in Q33

$$4T_1 - T_2 + 0 - T_4 = 30$$
$$-T_1 + 4T_2 - T_3 + 0 = 60$$
$$0 - T_2 + 4T_3 - T_4 = 70$$
$$-T_1 + 0 - T_3 + 4T_4 = 40$$

$$\begin{bmatrix} 4 & -1 & 0 & -1 & 30 \\ -1 & 4 & -1 & 0 & 60 \\ 0 & -1 & 4 & -1 & 70 \\ -1 & 0 & -1 & 4 & 40 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 0 & -1 & 4 & 40 \\ -1 & 4 & -1 & 0 & 60 \\ 0 & -1 & 4 & -1 & 70 \\ 4 & -1 & 0 & -1 & 30 \end{bmatrix} \sim r_1 \leftrightarrow r_4$$

$$\begin{bmatrix} -1 & 0 & -1 & 4 & 40 \\ 0 & 4 & 0 & -4 & 20 \\ 0 & -1 & 4 & -1 & 70 \\ 0 & -1 & -4 & 15 & 190 \end{bmatrix} \begin{matrix} \sim r_2 \leftarrow r_2 - r_1 \\ \sim r_4 \leftarrow r_4 + 4(r_1) \end{matrix}$$

$$\begin{bmatrix} +1 & 0 & +1 & -4 & -40 \\ 0 & 1 & 0 & -1 & 5 \\ 0 & -1 & 4 & -1 & 70 \\ 0 & -1 & -4 & 15 & 190 \end{bmatrix} \begin{matrix} \sim r_2 \leftarrow r_2\left(\frac{1}{4}\right) \\ \sim r_1 \leftarrow (-1) r_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & -4 & -40 \\ 0 & 1 & 0 & -1 & 5 \\ 0 & 0 & 4 & -2 & 75 \\ 0 & 0 & -4 & 14 & 195 \end{bmatrix} \begin{matrix} \sim r_3 \leftarrow r_2 + r_3 \\ \sim r_4 \leftarrow r_2 + r_4 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 1 & -4 & -40 \\ 0 & 1 & 0 & -1 & 5 \\ 0 & 0 & 4 & -2 & 75 \\ 0 & 0 & 0 & 12 & 270 \end{bmatrix} \sim r_4 \leftarrow r_3 + r_4$$

$$\left[\begin{array}{ccccc} 1 & 0 & 1 & -4 & -40 \\ 0 & 1 & 0 & -1 & 5 \\ 0 & 0 & 4 & -2 & 75 \\ 0 & 0 & 0 & 1 & 22.5 \end{array}\right] \sim r_4 \leftarrow \frac{1}{10} r_4$$

$T_1 + T_3 - 4T_4 = -40$ —— ①

$T_2 - T_4 = 5$ —— ②

$4T_3 - 2T_4 = 75$ —— ③

$\boxed{T_4 = 22.5}$ —— ④

put eq ④ in eq ③

$4T_3 - 2(22.5) = 75$

$4T_3 - 45 = 75$

$4T_3 = 75 + 45 = 120$

$\frac{\cancel{4}T_3}{\cancel{4}} = \frac{\cancel{120}^{30}}{4}$

$\boxed{T_3 = 30}$

put T_3 in eq ① and T_4 in eq ①

$T_1 + 30 - 4(22.5) = -40$

$T_1 - \cancel{1}60 = -40$

$T_1 = -40 + 60$

$\boxed{T_1 = 20}$

Solution (20, 27.5, 30, 22.5

put T_4 in eq ②

$T_2 - 22.5 = 5$

$T_2 = 22.5 + 5$

$\boxed{T_2 = 27.5}$

Exercise # 1.2

Q4

$$\begin{bmatrix} 1 & 3 & 5 & 7 \\ 3 & 5 & 7 & 9 \\ 5 & 7 & 9 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & +4 & +8 & +12 \\ 0 & +8 & +16 & +34 \end{bmatrix} \quad \begin{array}{l} \tilde{r}_2 \leftarrow 3r_1 - r_2 \\ \tilde{r}_3 \leftarrow 5r_1 + r_3 \end{array}$$

$$\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & 4 & 8 & 12 \\ 0 & 0 & 0 & 10 \end{bmatrix} \quad \tilde{r}_3 \leftarrow r_3 - 2r_2$$

$$\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{array}{l} \tilde{r}_2 \leftarrow \frac{1}{4}(r_2) \\ \tilde{r}_3 \leftarrow \left(\frac{1}{10}\right) r_3 \end{array}$$

$$\begin{bmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{array}{l} \tilde{r}_1 \leftarrow r_1 - 3r_2 \\ \tilde{r}_2 \leftarrow r_2 - 3r_3 \end{array}$$

$$\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \tilde{r}_1 \leftarrow r_1 + 2r_3$$

Pivot colums : 1, 2, 4

Q6 exercise: 5 for 3x2 matrix

$$\begin{bmatrix} \square & * \\ 0 & \square \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} \square & * \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & \square \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Q11

$$\begin{bmatrix} 3 & -4 & 2 & 0 \\ -9 & 12 & -6 & 0 \\ -6 & 8 & -4 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -4 & 2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} \tilde{r}_2 \leftarrow r_2 + 3r_1 \\ \tilde{r}_3 \leftarrow r_3 + 2r_1 \end{matrix}$$

$$\begin{bmatrix} 1 & -4/3 & 2/3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \tilde{r}_1 \leftarrow (\tfrac{1}{3})r_1$$

$$x_1 - \frac{4}{3}x_2 + \frac{2}{3}x_3 = 0$$

General solution: $\begin{cases} x_1 = \frac{4}{3}x_2 - \frac{2}{3}x_3 \\ x_2 \text{ is free} \\ x_3 \text{ is free} \end{cases}$

(a)

$$\begin{bmatrix} \square & * & * & * \\ 0 & \square & * & * \\ 0 & 0 & \square & 0 \end{bmatrix}$$

The system is consistent, with a unique solution.

(b)

$$\begin{bmatrix} 0 & \square & * & * & * \\ 0 & 0 & \square & * & * \\ 0 & 0 & 0 & 0 & \square \end{bmatrix}$$

The system is inconsistent.

Q20 $x_1 + 3x_2 = 2$

$3x_1 + hx_2 = K$

$$\begin{bmatrix} 1 & 3 & 2 \\ 3 & h & K \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 2 \\ 0 & h-9 & K-6 \end{bmatrix} \quad r_2 \leftarrow r_2 - 3r_1$$

(a) No solution: when $h=9$ and $K \neq 6$, the system is inconsistent. ~~and h~~

(b) A unique solution: when $h \neq 9$, the system is consistent and has a unique solution.

(c) Many solutions: when $h = 9$ and $K = 6$, the system is consistent and has many solutions.

Q23 Yes, the system is consistent because with three pivots, there must be a pivot in the last row of the coefficient matrix. The reduced echelon form can't contain a row of the form [0 0 0 0 0 1]

Q33 $P(t) = a_0 + a_1 t + a_2 t^2$

From given data in question, the augmented form will be,

$$\begin{bmatrix} 1 & 1 & 1 & 12 \\ 1 & 2 & 4 & 15 \\ 1 & 3 & 9 & 16 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 & 12 \\ 0 & 1 & 3 & 3 \\ 0 & 2 & 8 & 4 \end{bmatrix} \quad \begin{array}{l} \tilde{r}_2 \leftarrow r_2 - r_1 \\ \tilde{r}_3 \leftarrow r_3 - r_1 \end{array}$$

$$\begin{bmatrix} 1 & 1 & 1 & 12 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 2 & -2 \end{bmatrix} \quad \tilde{r}_3 \leftarrow r_3 - 2r_2$$

$$\begin{bmatrix} 1 & 1 & 1 & 12 \\ 0 & 1 & 3 & 3 \\ 0 & 0 & 1 & -1 \end{bmatrix} \quad \tilde{r}_3 \leftarrow \left(\frac{1}{2}\right) r_3$$

$$\begin{bmatrix} 1 & 1 & 0 & 13 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & -1 \end{bmatrix} \quad \begin{array}{l} \tilde{r}_1 \leftarrow r_1 - r_3 \\ \tilde{r}_2 \leftarrow r_2 - 3r_3 \end{array}$$

$$\begin{bmatrix} 1 & 0 & 0 & 7 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 1 & -1 \end{bmatrix} \quad \tilde{r}_1 \leftarrow r_1 - r_2$$

The polynomial is $p(t) = 7 + 6t - t^2$